ModelIt!

Teacher's Guide

Why Can't Humans Photosynthesize?

Modeling Energy Flow Between Photosynthesis and Cellular Respiration

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-LS1-7, HS-LS2-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-7, HS-LS2-5

Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules and oxygen molecules are broken and the bonds in new compounds are formed, resulting in a net transfer of energy. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to illustrate the energy transformations in photosynthesis and cellular respiration and the cycling of carbon through biological and geological systems.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms / LS2.B Cycles of Matter and Energy Transfer in Ecosystems** — Cellular respiration breaks glucose bonds to release energy for cellular work. Photosynthesis and respiration together cycle carbon among the biosphere, atmosphere, and geosphere.
Crosscutting Concept	**Energy and Matter** — Students track energy flow from sunlight through photosynthesis to glucose to cellular respiration to ATP, accounting for energy losses at each step and understanding why the cycle is not 100% efficient.

CAST Testing Alignment

MULTIPLE CHOICE:

If photosynthesis converts 1.5% of incoming solar energy to glucose and a human body has 1.7 m^2 of surface area receiving 500 W/m^2 of sunlight for 8 hours, how much glucose energy would a photosynthetic human produce per day?

CONSTRUCTED RESPONSE:

Using your model, explain why humans cannot photosynthesize by addressing: (1) the efficiency of photosynthesis, (2) the surface-area-to-volume constraint, (3) the energy demands of cellular respiration in humans, and (4) the complementary relationship between photosynthesis and respiration in the carbon cycle.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plants and animals have a deal: plants capture sunlight and store it in glucose, then animals eat the plants and burn that glucose for energy. This partnership — photosynthesis and cellular respiration — is the engine that drives virtually all life on Earth. But why can't animals cut out the middleman and photosynthesize for themselves? The answer reveals deep truths about energy, geometry, and the evolution of life.

The Efficiency Problem

Photosynthesis captures only about 1-2% of incoming solar energy as chemical energy in glucose. Of the sunlight hitting a leaf, about 50% is the wrong wavelength (chlorophyll absorbs mainly red and blue), 20% is reflected or transmitted, and most of the remaining energy is lost as heat during the chemical reactions. The theoretical maximum efficiency of photosynthesis is about 11%, but real plants achieve 1-2% under field conditions. After billions of years of evolution, this is still the best biology can do.

The Surface-Area-to-Volume Problem

Energy capture is proportional to surface area; energy consumption is proportional to volume. For a sphere, surface area scales as r^2 while volume scales as r^3. This means larger organisms have proportionally less surface area relative to their energy needs. A single-celled alga can meet ALL its energy needs with photosynthesis. A tree needs enormous flat leaves but has very low metabolic rate per unit mass. A human has too little surface area and too high a metabolic rate — the geometry is impossible.

The Numbers

Calculation: Maximum solar intensity at Earth's surface = 1,000 W/m^2. Average over a day (accounting for angle, weather, night) ≈ 250 W/m^2. Human surface area ≈ 1.7 m^2. Energy captured at 1.5% efficiency = 1.7 × 250 × 0.015 = 6.4 W = 5,500 kcal per day... but wait, that's watts of SOLAR energy. At 1.5% photosynthetic efficiency, usable glucose energy ≈ 60-100 kcal/day. A human needs 2,000 kcal/day. You'd need to be 20-30 times your current surface area to survive — you'd need to be a flat sheet about 6 meters across.

The Carbon Cycle Connection

Photosynthesis and cellular respiration are mirror-image reactions: $6CO_2 + 6H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6O_2$ (photosynthesis) and $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$ (respiration). Together, they cycle carbon between the atmosphere (as CO_2) and the biosphere (as organic molecules). Every carbon atom in your body was once CO_2 in the air, captured by a plant, and will return to CO_2 when that molecule is respired. This cycle has maintained atmospheric CO_2 balance for hundreds of millions of years — until humans started burning fossil fuels faster than photosynthesis can absorb them.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify sunlight intensity, chlorophyll concentration, glucose production rate, cellular respiration rate, and net energy output as the key components of the photosynthesis-respiration energy system.

E **Establish Relationships**

Students determine that sunlight and chlorophyll drive glucose production, but cellular respiration consumes glucose at a rate determined by body size and metabolic demand, and the net energy output determines whether an organism can survive on photosynthesis alone.

V **Visualize & Model**

Students build a computational model showing energy flow from sunlight through photosynthesis to glucose to cellular respiration, calculating net energy balance for organisms of different sizes.

E **Evaluate Outcomes**

Students run optimal plant, photosynthetic human, and scaling problem scenarios to discover the surface-area-to-volume constraint and calculate why photosynthetic humans are mathematically impossible.

R **Revise & Extend**

Students add CO2 concentration, leaf area index, or metabolic efficiency to model more complete energy balance and carbon cycling dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"A leaf turns sunlight into food using nothing but water and air. Why can't you? Imagine never having to eat again — just stand in the sun."*

Watch For

Ask: If you could photosynthesize, would you? How much time in the sun would you need? Let students speculate.

Slide 2: Learning Objectives

3 min

Say: *"Today we're running the numbers on a wild idea — and discovering why evolution chose eating over photosynthesizing for animals."*

Watch For

Have students read objectives. Pre-teach 'photosynthesis' and 'cellular respiration' as mirror reactions.

Slide 3: Big Question

3 min

Say: *"Plants do it. Some sea slugs do it. Some salamander embryos sort of do it. So what's stopping us?"*

Watch For

Students write their best hypothesis for why humans can't photosynthesize. Collect for later revisiting.

Slide 4: LEVER Framework

2 min

Say: *"We're building an energy budget model — calculating exactly how much energy photosynthesis could give a human versus how much a human needs."*

Watch For

Preview LEVER steps. Emphasize this is a QUANTITATIVE question — the answer is in the numbers.

Slide 5: Activity 1: Components

8 min

Say: *"What factors determine whether an organism can survive on photosynthesis? Which ones can biology control?"*

Watch For

Guide sorting. Sunlight and chlorophyll are the inputs; glucose production and respiration rate determine the balance.

Slide 6: Activity 2: Connections

8 min

Say: *"Sunlight hits chlorophyll. Chlorophyll makes glucose. Your cells burn glucose for ATP. But at EVERY step, energy is lost. How much?"*

Watch For

Build the energy cascade: 1000 W/m² solar → ~15 W/m² captured → glucose → ATP → work. Each arrow loses energy.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's run the math. Cover your entire body with chlorophyll. Stand in the sun all day. How many calories do you produce?"*

Watch For

Students calculate photosynthetic output for 1.7 m² at 1.5% efficiency. Compare to 2,000 kcal daily need. Watch the deficit.

Slide 8: Discoveries

5 min

Say: *"About 60-100 calories from photosynthesis. You need 2,000. You'd have to be a flat green sheet 6 meters across. Evolution chose eating for a reason."*

Watch For

Revisit initial hypotheses. Most students will not have guessed the surface-area-to-volume problem. Discuss body plan evolution.

Slide 9: STEM Challenge

12 min

Say: *"Natural photosynthesis is only 1-2% efficient. Silicon solar cells hit 25%. Can we build something biological that does better?"*

Watch For

Groups design bio-solar systems — improved organisms, artificial photosynthesis, or bio-inspired materials. Present and evaluate.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sunlight Intensity, Chlorophyll Concentration
- **INTERNAL:** Glucose Production Rate, Cellular Respiration Rate, Net Energy Output

Relationships

- **Sunlight Intensity to Glucose Production Rate:** POSITIVE (+) up to saturation — Higher sunlight intensity drives more photosynthetic reactions, increasing glucose production — but only up to a saturation point where the photosynthetic machinery is running at maximum capacity. Beyond this point, additional light cannot be used.
- **Chlorophyll Concentration to Glucose Production Rate:** POSITIVE (+) — Higher chlorophyll concentration allows more light photons to be captured and channeled into the light-dependent reactions, increasing the overall rate of glucose production. This is why leaves are densely packed with chloroplasts.
- **Glucose Production Rate to Net Energy Output:** POSITIVE (+) — More glucose produced by photosynthesis increases the net energy available. Net Energy Output = glucose energy from photosynthesis - glucose energy consumed by respiration. When production exceeds consumption, the organism grows; when consumption exceeds production, it starves.

Key Discoveries

- Photosynthesis is remarkably inefficient at converting sunlight to chemical energy — only about 1-2% of incoming solar energy ends up stored in glucose molecules
- A photosynthetic human would produce only about 60-100 kcal per day from their skin surface area — roughly 3-5% of the 2,000 kcal needed for survival
- The surface-area-to-volume ratio is the fundamental constraint — as organisms get larger, their energy needs (proportional to volume) grow much faster than their ability to capture light (proportional to surface area)
- Photosynthesis and cellular respiration are complementary reactions that cycle carbon between organisms and the atmosphere — the glucose and O2 produced by one are consumed by the other

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants get their mass from soil"	This is one of the oldest misconceptions in biology, dating back to ancient Greece. In reality, the vast majority of a plant's dry mass comes from CO2 in the air, not minerals from soil. Photosynthesis fixes atmospheric carbon into glucose, which is then used to build cellulose, lignin, and all other organic molecules. A tree is literally made of air and water. Soil provides only mineral nutrients (nitrogen, phosphorus, potassium) that make up a tiny fraction of plant mass.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Photosynthesis and respiration are opposites that cancel out"	While the CHEMICAL equations are mirror images (one produces glucose + O2, the other consumes them), they serve completely different biological functions and occur in different organelles. Photosynthesis CAPTURES and STORES solar energy in chemical bonds. Respiration RELEASES that stored energy for cellular work. A plant does BOTH simultaneously — photosynthesis in chloroplasts and respiration in mitochondria. During the day, photosynthesis exceeds respiration (net O2 producer). At night, only respiration occurs (net CO2 producer).

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If humans had chlorophyll, we wouldn't need to eat"	Even with perfect chlorophyll coverage, a human body's 1.7 m^2 surface area at 1.5% photosynthetic efficiency would produce only 60-100 kcal per day — roughly 3-5% of the 2,000 kcal daily requirement. The fundamental problem is geometry: our metabolic needs (proportional to body volume) vastly exceed our light-capture capacity (proportional to surface area). You would need to be a flat green sheet about 6 meters in diameter to survive on photosynthesis alone.